Noisy continuous sensing of a Sobolev ellipsoid: a uniform upper bound

Matthew J. Colbrook*

Department of Applied Mathematics and Theoretical Physics
University of Cambridge, Cambridge CB3 0WA, United Kingdom
m.colbrook@damtp.cam.ac.uk

4 November 2026

Abstract

We construct adaptive continuous noisy measurements for the Sobolev ellipsoid in AIM 639. Writing $L = \log_2(1/\delta)$, the bound is $e_{n,\delta} \le 256/[L \log_2(1 + 2^n/L)]$. It matches known lower bounds in two parameter regimes. A transition gap remains, so this is partial progress on the joint asymptotic target.

Catalogue target. AIM 639 at repository revision `c4036871b0bf`. The package preserves its full statement in `statement.md`; the review and target comparison are linked from the package README.

1 Problem 639: a uniform upper bound, with a transition gap

This section is a partial result. It does not determine the joint order throughout the full (n, δ) range. Let $D : \ell_2 \to \ell_2$ be given by $(Dx)_j = x_j/j$, with input the closed unit ball, and let

$$L = \log_2(1/\delta) \ge 1. \tag{639.1}$$

Measurements take values in $[-1, 1]$, are continuous in the input for each fixed history, and receive adversarial errors of magnitude at most δ.

Theorem 1.1 (A quantitative partial result)**.** *For every integer $m \ge 1$, there is an admissible algorithm with error at most $4/m$ using no more than*

$$\lceil \log_2(m + 1) \rceil + \left\lceil \frac{10m}{L} \right\rceil \tag{639.2}$$

measurements. Consequently,

$$e_n(\delta) \le \frac{256}{L \log_2(1 + 2^n/L)} \qquad (n \ge 1,\ 0 < \delta \le 1/2). \tag{639.3}$$

The colored-grid construction for exact measurements originates in [1, Algorithm 2]; the robust binary selection principle is related to [2, Theorem 3]. The additional ingredients here are robust finite-alphabet cell coding, a quantitative cell count for the Sobolev ellipsoid, and the uniform inversion of the resulting budget.

*The solutions were found by AI. The submitted arguments and the repository audit are AI work; independent human review and formal verification are not claimed.

1.1 A robust encoder for a colored cover

Lemma 1.2. *Let E be compact metric, with a finite open cover $U_1, \dots, U_N$ whose members have diameter at most ρ. Suppose they can be colored with q colors so that distinct members of the same color are disjoint. For $0 < \delta < 1$, put $M = \lceil 1/\delta \rceil \geq 2$. There is a continuous adaptive measurement algorithm, with adversarial noise δ, which recovers a point of E to error at most ρ using*

$$\lceil \log_2 q \rceil + \lceil \log_M N \rceil \tag{C1}$$

queries with values in $[-1, 1]$.

Proof. For a fixed previous noisy history h, let C_h be the feasible input set. It is compact: each previously fixed branch functional is continuous, and its noise-admissibility inequality is closed. Maintain a set I of candidate colors whose cover sets cover C_h. Initially, all colors are candidates.

Split I into two parts I_0, I_1, each of size at most $\lceil |I| /2 \rceil$, and let V_b be the union of the cover members of colors in I_b. The compact sets

$$F_0 = C_h \setminus V_0, \qquad F_1 = C_h \setminus V_1 \tag{C2}$$

are disjoint. Choose a continuous $\lambda : E \to [-1, 1]$ that is -1 on F_0 and $+1$ on F_1; a suitable constant handles an empty set. On an answer $y \geq 0$, retain I_0; on $y < 0$, retain I_1. If an input lay outside the retained union, its exact answer would be respectively -1 or $+1$, incompatible with $|y - \lambda(x)| \leq \delta < 1$. Thus the covering invariant persists, and after $\lceil \log_2 q \rceil$ queries one color remains.

The surviving cover members are pairwise disjoint and cover C_h. Their intersections with C_h are finitely many clopen pieces of C_h, hence are compact. Choose M equally spaced levels in $[-1, 1]$. Their separation is

$$\frac{2}{M - 1} > 2\delta, \tag{C3}$$

because $M - 1 < 1/\delta$. Encode the cell indices by M-ary strings of length $\lceil \log_M N \rceil$. For each digit, the assigned level is continuous on C_h, being constant on its finitely many clopen pieces. Tietze extension gives a continuous function on E with values in $[-1, 1]$. Every admissible noisy digit answer determines its unique level, so all digits recover the cell index. Return a fixed point of that cover member. Empty feasible branches may use arbitrary constant queries. No continuity in the previous answers is needed. □

1.2 A cover with exponentially many cells and $m + 1$ colors

Consider the finite-dimensional ellipsoid

$$E_m = \left\{ z \in \mathbb{R}^m : \sum_{j=1}^{m} j^2 z_j^2 \leq 1 \right\}. \tag{639.4}$$

Set $s = 1/(m\sqrt{m})$, and use the cubical grid $s\mathbb{Z}^m$. For $k = 0, \dots, m$, let F_k be the union of its k-dimensional faces, and put

$$\tau_k = \frac{m + 1 - k}{4(m + 1)}, \qquad H_k = \{ z : \operatorname{dist}_\infty(z, F_k) \leq \tau_k s \}. \tag{639.5}$$

The sets

$$C_0 = H_0, \qquad C_k = H_k \setminus \bigcup_{l<k} H_l \tag{639.6}$$

partition $\mathbb{R}^m$, since $H_m = \mathbb{R}^m$.

For $k \ge 1$, a point of C_k has exactly $m-k$ coordinates within $\tau_k s$ of a grid hyperplane, while the remaining k coordinates are farther than $\tau_{k-1}s$ from the grid. It therefore has a unique associated k-face: the first coordinates specify its fixed integer coordinates, and the others specify their open unit grid intervals. Partition C_k by these associated faces; for $k=0$, use the unique nearby grid vertex. Take closures of all resulting cells.

Each closed cell has ℓ_∞ diameter at most s, and therefore Euclidean diameter at most $1/m$. Distinct cells of the same k have ℓ_∞ separation at least

$$g = \frac{s}{4(m+1)}. \tag{639.7}$$

To verify this, if their fixed/free coordinate patterns differ, some coordinate is within $\tau_k s$ of the grid for one cell and at least $\tau_{k-1}s$ away for the other. If the patterns agree, either a fixed integer or a free grid interval differs, giving at least that separation. At $k=0$, the separation is at least $s/2$.

Enlarge each cell by its open ℓ_∞ neighborhood of radius $g/3$, intersect it with E_m, and retain the nonempty sets. They form an open cover of E_m, with colors $k=0,\dots,m$. Same-color sets remain disjoint, and each has Euclidean diameter less than $2/m$.

The number N of retained cells is at most 2^{10m}. Associate each cell with a vertex of its grid face. If its enlargement meets E_m, that vertex lies within ℓ_∞ distance $2s$ of E_m. The disjoint half-open cubes of side s attached to these vertices lie in

$$E_m + 3s[-1,1]^m \subset E_m + \frac{3}{m}B_2^m \subset 4E_m, \tag{639.8}$$

because E_m contains $m^{-1}B_2^m$. There are at most 3^m incident grid faces per vertex, and $\operatorname{vol}(E_m) = \omega_m/m!$, where ω_m is the volume of the Euclidean unit ball. Hence

$$\begin{aligned} N &\le 3^m \frac{4^m \omega_m}{m!\, s^m} \\ &\le \left(12e\sqrt{2\pi e}\right)^m < 2^{8m} \le 2^{10m}. \end{aligned} \tag{639.9}$$

Here we used $\omega_m \le (2\pi e/m)^{m/2}$ and $m! \ge (m/e)^m$. All constants are independent of m.

1.3 The sensing algorithm and its budget

Proof of Theorem 1.1. Apply Lemma 1.2 to the cover just constructed. Compose each query with the continuous map

$$x \longmapsto (x_1, x_2/2, \dots, x_m/m) \in E_m. \tag{639.10}$$

Since $\log_2 M \ge L$, the number of measurements is at most (639.2), and the first m output coordinates are recovered within $2/m$. Return zero in the tail, for which

$$\sum_{j>m} \frac{x_j^2}{j^2} \le \frac{1}{(m+1)^2}. \tag{639.11}$$

The total error is less than $3/m$, and hence at most $4/m$. All queries are continuous on the original unit ball, take values in $[-1,1]$, and handle every admissible adversarial answer.

For the explicit inversion, put

$$Z = L\log_2(1 + 2^n/L). \tag{639.12}$$

If $Z < 64$, the zero-output algorithm has error at most $1 \le 64/Z$. Otherwise choose $m = \lfloor Z/32 \rfloor$, so $m \ge Z/64$. The budget in (639.2) is bounded by

$$\log_2 m + 3 + \frac{10m}{L}. \tag{639.13}$$

Write $t = 2^n/L$ and $u = \log_2(1+t)$, so $Z = Lu$. Using $m \le Lu/32$, (639.13) is at most

$$n - 2 + \log_2(u/t) + \frac{5}{16}u. \tag{639.14}$$

The last two terms are less than 2 for every $t > 0$. For $t \le 1$, use $u \le t/\log 2$ and $u \le 1$. For $t \ge 1$, write $v = \log_2 t \ge 0$ and $u \le 1 + v$; the terms are bounded by

$$\frac{5}{16} + \log_2(1+v) - \frac{11}{16}v, \tag{639.15}$$

whose maximum is less than 2, as follows by differentiation. Thus the integer budget is at most n, and the error is at most

$$\frac{4}{m} \le \frac{256}{Z}. \tag{639.16}$$

This proves (639.3). □

1.4 What the result does and does not settle

Combining (639.3) with the entropy lower bound [2] and the antipodal lower bound recorded in the catalogue via the adversarial-transcript argument of [3] gives

$$c \max\left\{2^{-n}, \frac{1}{nL}\right\} \le e_n(\delta) \le \frac{256}{L \log_2(1 + 2^n/L)}. \tag{639.17}$$

These estimates match up to absolute constants in two regimes:

$$\begin{array}{c|c} L \le 2^{n/2} & e_n(\delta) \asymp (nL)^{-1} \\ L \ge 2^n & e_n(\delta) \asymp 2^{-n}. \end{array} \tag{639.18}$$

For example, in the first regime $\log_2(1 + 2^n/L)$ lies between $n/2$ and $n+1$; in the second it is comparable to $2^n/L$. The intermediate transition is not determined. At L of order $2^n/n$, the remaining ratio can be of order $n/\log n$. Neither the cover construction nor the two separate lower bounds justify declaring Problem 639 solved.

References

[1] D. Krieg, E. Novak and M. Ullrich, *How many continuous measurements are needed to learn a vector?*, arXiv:2412.06468v2, 22 September 2026. https://arxiv.org/abs/2412.06468.

[2] D. Krieg, E. Novak, L. Plaskota and M. Ullrich, *Noisy nonlinear information and entropy numbers*, J. Fourier Anal. Appl. **32** (2026), 36. https://arxiv.org/abs/2510.23213; https://doi.org/10.1007/s00041-026-10249-z.

[3] S. Sadeghi, *Continuous adaptive measurements: a sharp asymptotic bound*, AIM 553 research submission, 4 October 2026, Section 1.3. Pinned source; see also the pinned catalogue statement of AIM 639 for the ellipsoid restriction.